Volatility models for REMX: GARCH, GARCH-X, GARCHAND, GARCHND and EGARCH-X

Summary of all models estimated in the notebooks **03-GP-models-gaussian**, **03-GP-models-student** and **04-extension-models**. Each section contains the variance expression, a brief description of what the model is meant to demonstrate, and the main estimation results (Gaussian first, Student-t second). The *Sig.* column summarises significance: *** = 1%, ** = 5%, * = 10%. A final part covers the six robustness checks from notebooks **06-robustness-checks** and **07-extra-robustness-checks**.

1. GARCH(1,1) — Baseline

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1}$$

What we want to demonstrate: Canonical Bollerslev (1986) specification. It serves as the benchmark against which every sentiment-augmented model is compared. The goal is to confirm persistence ($\alpha+\beta$ close to 1) and that the basic parameters are significant; all later extensions are evaluated relative to this fit.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0857	0.0301	2.8497	0.0044	***
alpha[1]	0.0893	0.0161	5.5536	0.0000	***
beta[1]	0.8975	0.0182	49.2043	0.0000	***

```
Log-likelihood : -5982.3069
AIC : 11970.6138
BIC : 11988.3882
alpha[1] + beta[1] = 0.9867 (stationary (<1))
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0878	0.0299	2.9308	0.0034	***
alpha[1]	0.0841	0.0150	5.5880	0.0000	***
beta[1]	0.9010	0.0178	50.6845	0.0000	***
nu	11.1674	2.3239	4.8054	0.0000	***

```
Log-likelihood : -5956.2330
AIC : 11920.4660
BIC : 11944.1652
alpha[1] + beta[1] = 0.9850 (stationary (<1))
nu = 11.1674 (smaller -> heavier tails; Gaussian as nu -> infinity)
```

2. GARCH-X (tone) — G&P Eq. (4)

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1} + \gamma \cdot \text{tone}^2_{t-1}$$

What we want to demonstrate: Symmetric extension that injects the daily average news tone as an exogenous regressor in the variance equation. We want to test whether news intensity (regardless of sign) generates additional volatility in REMX, i.e. whether γ is statistically significant.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0907	0.0305	2.9703	0.0030	***
alpha	0.0898	0.0162	5.5564	0.0000	***
beta	0.8967	0.0184	48.8103	0.0000	***
gamma	-0.0047	0.0151	-0.3121	0.7549	

```
Log-likelihood : -5982.4884
AIC : 11972.9767
BIC : 11996.6759
alpha + beta : 0.9866 (stationary (<1))
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0953	0.0313	3.0451	0.0023	***
alpha	0.0844	0.0151	5.6021	0.0000	***
beta	0.9005	0.0178	50.6344	0.0000	***
gamma	-0.0078	0.0160	-0.4866	0.6266	
nu	11.0876	2.2973	4.8265	0.0000	***

```
Log-likelihood : -5956.1902
AIC : 11922.3804
BIC : 11952.0044
alpha + beta : 0.9849 (stationary (<1))
Estimated degrees of freedom: nu_hat = 11.088
Note: t-statistic on nu vs 0 is not economically meaningful (nu > 2 by construction).
```

3. GARCH-X (log article growth) — G&P Eq. (4)

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1} + \gamma \cdot \text{log\_art\_growth}^2_{t-1}$$

What we want to demonstrate: Same structure as the previous GARCH-X, but using the log-growth in article volume ($\log(1+N_t) - \log(1+N_{t-1})$) as a proxy for news intensity. The log transform replaces the raw growth rate to remove a single-day outlier (art_growth = 61 on 2015-10-23) that would otherwise dominate the squared term. We want to see whether sustained changes in media coverage (a significant γ) anticipate next-day volatility.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0780	0.0370	2.1088	0.0350	**
alpha	0.0908	0.0160	5.6643	0.0000	***
beta	0.8961	0.0181	49.5127	0.0000	***
gamma	0.0170	0.0419	0.4068	0.6842	

```
Log-likelihood : -5982.4084
AIC : 11972.8167
BIC : 11996.5159
alpha + beta : 0.9869 (stationary (<1))
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0894	0.0358	2.4988	0.0125	**
alpha	0.0852	0.0149	5.7116	0.0000	***
beta	0.8996	0.0176	51.0750	0.0000	***
gamma	-0.0003	0.0330	-0.0077	0.9939	
nu	11.1163	2.3046	4.8235	0.0000	***

```
Log-likelihood : -5956.3184
AIC : 11922.6368
BIC : 11952.2608
alpha + beta : 0.9848 (stationary (<1))
Estimated degrees of freedom: nu_hat = 11.116
Note: t-statistic on nu vs 0 is not economically meaningful (nu > 2 by construction).
```

4. GARCHAND (tone) — G&P Eq. (5)

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1} + \gamma\,(1 + d_1 \cdot \theta) \cdot \mathrm{tone}^2_{t-1},\ d_1 = 1\{\mathrm{tone}_{t-1} < 0\}$$

What we want to demonstrate: Asymmetric version of GARCH-X: when the previous day's tone is negative, its impact on variance is amplified by the factor $(1 + \theta)$. We want to demonstrate that negative news drives a stronger volatility response than positive news ($\theta > 0$ and significant).

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0899	0.0319	2.8172	0.0048	***
alpha	0.0899	0.0160	5.6072	0.0000	***
beta	0.8967	0.0181	49.4677	0.0000	***
gamma	-0.0030	0.0162	-0.1857	0.8527	
theta	0.4971	7.5039	0.0663	0.9472	

```
Log-likelihood : -5982.4997
AIC : 11974.9994
BIC : 12004.6234
alpha + beta : 0.9866 (stationary (<1))
Fraction of days with tone_{t-1} < 0 (d1 = 1): 0.390
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0947	0.0307	3.0788	0.0021	***
alpha	0.0844	0.0151	5.6078	0.0000	***
beta	0.9005	0.0178	50.6732	0.0000	***
gamma	-0.0057	0.0165	-0.3462	0.7292	
theta	0.4959	4.0394	0.1228	0.9023	
nu	11.0871	2.2980	4.8246	0.0000	***

```
Log-likelihood : -5956.1746
AIC : 11924.3493
BIC : 11959.8980
alpha + beta : 0.9849 (stationary (<1))
Fraction of days with tone_{t-1} < 0 (d1 = 1): 0.390
Estimated degrees of freedom: nu_hat = 11.087
```

5. GARCHAND (log article growth) — G&P Eq. (6)

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1} + \gamma \cdot d_2 \cdot \text{log\_art\_growth}^2_{t-1},\ d_2 = 1\{\text{log\_art\_growth}_{t-1} > 0\}$$

What we want to demonstrate: The article-volume effect is switched on only on days in which the news flow GROWS relative to the previous day (log_art_growth > 0). We want to show that novelty (rising coverage) is what moves volatility, rather than low news volumes.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0805	0.0369	2.1798	0.0293	**
alpha	0.0903	0.0159	5.6691	0.0000	***
beta	0.8964	0.0181	49.4928	0.0000	***
gamma	0.0272	0.0936	0.2904	0.7715	

```
Log-likelihood : -5982.4728
AIC : 11972.9457
BIC : 11996.6449
alpha + beta : 0.9866 (stationary (<1))
Fraction of days with log_art_growth_{t-1} > 0 (d2 = 1): 0.473
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0938	0.0383	2.4516	0.0142	**
alpha	0.0875	0.0156	5.6277	0.0000	***
beta	0.8994	0.0181	49.6810	0.0000	***
gamma	0.0002	0.0850	0.0022	0.9983	
nu	8.0077	0.7625	10.5017	0.0000	***

```
Log-likelihood : -5958.4131
AIC : 11926.8262
BIC : 11956.4501
alpha + beta : 0.9869 (stationary (<1))
Fraction of days with log_art_growth_{t-1} > 0 (d2 = 1): 0.473
Estimated degrees of freedom: nu_hat = 8.008
```

6. GARCHND (tone & log article growth) — G&P Eq. (7)

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1} + \gamma \cdot d_3 \cdot x^2_{t-1},\ d_3 = 1\{\sigma^2_{t-1} \geq \kappa\},\ x \in \{\text{tone, log\_art\_growth}\}$$

What we want to demonstrate: The news impact activates only when the previous day's variance exceeds a threshold κ, calibrated from an annualised volatility level (30% and 50%). We want to test whether the sentiment effect is non-linear in the volatility regime: a significant γ in the high-volatility regime would imply that news amplifies volatility only when the market is already stressed.

Gaussian

GARCHND x = Tone kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0929	0.0332	2.8029	0.0051	***
alpha	0.0908	0.0173	5.2433	0.0000	***
beta	0.8920	0.0208	42.9489	0.0000	***
gamma	0.0213	0.0239	0.8914	0.3727	

```
Log-likelihood : -5982.0113
AIC : 11972.0227
BIC : 11995.7219
alpha + beta : 0.9828 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.616
```

GARCHND x = Tone kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0412	0.0243	1.6921	0.0906	*
alpha	0.0845	0.0077	11.0197	0.0000	***
beta	0.9162	0.0086	106.8527	0.0000	***
gamma	-1.2195	0.1779	-6.8566	0.0000	***

```
Log-likelihood : -5973.4457
AIC : 11954.8915
BIC : 11978.5907
alpha + beta : 1.0008 (non-stationary (>=1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.059
```

GARCHND x = log_art_growth kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.1093	0.0329	3.3242	0.0009	***
alpha	0.0872	0.0117	7.4328	0.0000	***
beta	0.8818	0.0218	40.4393	0.0000	***
gamma	0.2302	0.1017	2.2641	0.0236	**

```
Log-likelihood : -5979.0497
AIC : 11966.0994
```

```
BIC : 11989.7986
alpha + beta : 0.9690 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.634
```

GARCHND x = log_art_growth kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0565	0.0317	1.7793	0.0752	*
alpha	0.0927	0.0093	9.9933	0.0000	***
beta	0.9051	0.0166	54.4964	0.0000	***
gamma	-1.2088	0.1894	-6.3837	0.0000	***

```
Log-likelihood : -5979.0803
AIC : 11966.1606
BIC : 11989.8598
alpha + beta : 0.9977 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.075
```

Student-t

GARCHND x = Tone kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0919	0.0304	3.0267	0.0025	***
alpha	0.0852	0.0146	5.8442	0.0000	***
beta	0.8977	0.0181	49.6099	0.0000	***
gamma	0.0114	0.0215	0.5333	0.5938	
nu	11.1736	2.3005	4.8571	0.0000	***

```
Log-likelihood : -5956.1755
AIC : 11922.3509
BIC : 11951.9749
alpha + beta : 0.9829 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.614
Estimated degrees of freedom: nu_hat = 11.174
```

GARCHND x = Tone kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0421	0.0191	2.2089	0.0272	**
alpha	0.0614	0.0049	12.5824	0.0000	***
beta	0.9331	0.0097	96.3702	0.0000	***
gamma	-0.8868	0.0679	-13.0521	0.0000	***
nu	11.9034	2.4404	4.8777	0.0000	***

```
Log-likelihood : -5951.4249
AIC : 11912.8498
BIC : 11942.4738
```

```
alpha + beta : 0.9944 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.046
Estimated degrees of freedom: nu_hat = 11.903
```

GARCHND x = log_art_growth kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0999	0.0314	3.1761	0.0015	***
alpha	0.0845	0.0140	6.0252	0.0000	***
beta	0.8940	0.0170	52.7059	0.0000	***
gamma	0.0958	0.0617	1.5524	0.1206	
nu	11.1160	2.2781	4.8796	0.0000	***

```
Log-likelihood : -5955.5857
AIC : 11921.1715
BIC : 11950.7954
alpha + beta : 0.9785 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.633
Estimated degrees of freedom: nu_hat = 11.116
```

GARCHND x = log_art_growth kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0605	0.0346	1.7519	0.0798	*
alpha	0.0794	0.0113	7.0419	0.0000	***
beta	0.9135	0.0191	47.7192	0.0000	***
gamma	-1.1900	0.2009	-5.9233	0.0000	***
nu	11.3988	2.3566	4.8371	0.0000	***

```
Log-likelihood : -5953.9383
AIC : 11917.8766
BIC : 11947.5006
alpha + beta : 0.9928 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.060
Estimated degrees of freedom: nu_hat = 11.399
```

7. EGARCH(1,1)-X — Extension model

Conditional variance:

$$\ln(\sigma^2_t) = \omega + \alpha\,[|z_{t-1}| - E|z_{t-1}|] + \xi \cdot z_{t-1} + \beta \cdot \ln(\sigma^2_{t-1}) + \gamma_1 \cdot \text{tone}_{t-1} + \gamma_2 \cdot \text{log\_art\_growth}_{t-1}$$

What we want to demonstrate: Log-variance specification (Nelson, 1991) with two exogenous regressors. It (i) guarantees positivity of σ^2 without parameter restrictions, (ii) captures asymmetry through the leverage term ξ (negative shocks weigh more than positive ones), and (iii) allows tone and log_art_growth to enter linearly (not as squares). We want to show that EGARCH-X captures asymmetry more cleanly than GARCHAND, and to test whether γ_1 and γ_2 are significant in log-variance rather than in squared variance.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
mu	-0.0235	0.0368	-0.6398	0.5223	
omega	0.0306	0.0107	2.8504	0.0044	***
alpha	0.1726	0.0287	6.0075	0.0000	***
xi	-0.0231	0.0131	-1.7711	0.0765	*
beta	0.9814	0.0067	147.2149	0.0000	***
gamma_tone	0.0058	0.0068	0.8514	0.3946	
gamma_log_artg	-0.0414	0.0600	-0.6912	0.4894	

```
Log-likelihood : -5978.5917
AIC : 11971.1835
BIC : 12012.6571
beta = 0.9814 (stationary (|beta|<1))
Half-life of a log-variance shock: 37.0 trading days
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
mu	-0.0206	0.0365	-0.5642	0.5726	
omega	0.0244	0.0089	2.7469	0.0060	***
alpha	0.1600	0.0245	6.5395	0.0000	***
xi	-0.0160	0.0106	-1.5081	0.1315	
beta	0.9825	0.0058	168.0003	0.0000	***
gamma_tone	0.0084	0.0054	1.5553	0.1199	
gamma_log_artg	-0.0605	0.0585	-1.0354	0.3005	
nu	10.9830	2.2009	4.9901	0.0000	***

```
Log-likelihood : -5952.0206
AIC : 11920.0411
BIC : 11967.4395
beta = 0.9825 (stationary (|beta|<1))
Half-life of a log-variance shock: 39.4 trading days
```

nu = 10.9830 (Student-t dof; lower -> heavier tails. Note: the t-ratio for nu tests H0: nu = 0, which is not the meaningful null. To test Gaussian vs Student-t use an LR test against the Gaussian EGARCH-X.)

Part II — Robustness Checks (Notebooks 06 & 07)

Six diagnostic checks applied to the GARCHND κ=50% Student-t models (the best in-sample specifications). Checks 1–4 are from notebook 06; Checks 5–6 from notebook 07. The picture that emerges is more nuanced than the original static-OOS reading: the rolling-window forecast (Check 1, 813 origins with daily refit and two RV proxies) shows that both GARCHND specifications add genuine one-day-ahead predictive content over GARCH(1,1) — significant under both Diebold–Mariano and Clark–West — with the tone-augmented model retaining a significant edge at h = 5. The advantage disappears at h = 22, consistent with a short-lived news shock. The remaining checks qualify the in-sample identification: the news term is partly redundant with a regime intercept (Check 2), the t-ratio on γ is sensitive to the sigmoid sharpness (Check 3), the high-volatility regime is concentrated in a handful of long episodes (Check 4), and a permissive GDELT filter dominates the sentiment series (Check 6) — but the OOS gain at short horizons survives all of these qualifications and reverses the previous "regime artefact" verdict.

Check 1 — Rolling-window out-of-sample forecasting (h = 1, 5, 22 days)

Out-of-sample window: **2023-01-03 to 2026-03-31** (813 forecast origins). Parameters are re-estimated **daily** on a **1,260-day rolling sample**. The forecast at origin τ is the average of the iterated conditional-variance path $\sigma^2_{\tau+1|\tau},\dots,\sigma^2_{\tau+h|\tau}$; the target is the corresponding h-day mean of the realised-variance proxy. Two proxies are reported in parallel: **Parkinson** (RV_park) and **Rogers–Satchell** (RV_rs), both in $\%^2$/day. Losses: HRMSE, QLIKE (Patton 2011, robust to RV-noise), RMSE, MAE. Tests against the GARCH(1,1) Student-t baseline use Newey–West LRV at lag h–1: **Diebold–Mariano** (two-sided on squared-loss differential; DM > 0 $\Rightarrow$ candidate has lower MSE) and **Clark–West** (one-sided, the appropriate test for nested models since the baseline is γ = 0; CW > 0 $\Rightarrow$ candidate adds genuine predictive content). Rows sorted by model $\rightarrow$ proxy $\rightarrow$ h.

Loss functions (averaged over the 813 origins)

Model	Proxy	h	HRMSE	QLIKE	RMSE	MAE	N
GARCH(1,1)	Parkinson	1	5.4023	2.0760	4.8866	3.9585	813
GARCH(1,1)	Parkinson	5	2.9753	2.0933	4.4833	3.7785	809
GARCH(1,1)	Parkinson	22	2.6582	2.1211	4.1826	3.6778	792
GARCH(1,1)	Rogers-Satchell	1	12.5808	2.0617	4.8178	3.9280	813
GARCH(1,1)	Rogers-Satchell	5	2.9436	2.0791	4.3905	3.7677	809
GARCH(1,1)	Rogers-Satchell	22	2.6068	2.1039	4.1738	3.6484	792
GARCHND tone	Parkinson	1	5.3128	2.0730	4.7686	3.9090	813
GARCHND tone	Parkinson	5	2.9460	2.0928	4.3911	3.7591	809
GARCHND tone	Parkinson	22	2.7032	2.1348	4.2540	3.7962	792
GARCHND tone	Rogers-Satchell	1	12.5734	2.0580	4.6975	3.8701	813
GARCHND tone	Rogers-Satchell	5	2.9023	2.0779	4.2912	3.7452	809
GARCHND tone	Rogers-Satchell	22	2.6578	2.1174	4.2456	3.7671	792
GARCHND log_artg	Parkinson	1	5.3400	2.0750	4.7698	3.9104	813
GARCHND log_artg	Parkinson	5	2.9620	2.0949	4.4162	3.7639	809

GARCHND log_artg	Parkinson	22	2.6966	2.1361	4.3428	3.8460	792
GARCHND log_artg	Rogers-Satchell	1	12.4528	2.0599	4.6937	3.8731	813
GARCHND log_artg	Rogers-Satchell	5	2.9226	2.0799	4.3137	3.7527	809
GARCHND log_artg	Rogers-Satchell	22	2.6487	2.1187	4.3356	3.8295	792

Diebold–Mariano and Clark–West vs GARCH(1,1) baseline

Candidate	Proxy	h	DM	DM p	CW	CW p
GARCHND tone	Parkinson	1	+5.332	0.0000	+5.695	0.0000
GARCHND tone	Parkinson	5	+2.071	0.0383	+2.293	0.0109
GARCHND tone	Parkinson	22	–0.988	0.3230	–0.699	0.7577
GARCHND tone	Rogers-Satchell	1	+5.449	0.0000	+5.816	0.0000
GARCHND tone	Rogers-Satchell	5	+2.120	0.0340	+2.332	0.0098
GARCHND tone	Rogers-Satchell	22	–0.973	0.3305	–0.690	0.7549
GARCHND log_artg	Parkinson	1	+3.814	0.0001	+4.350	0.0000
GARCHND log_artg	Parkinson	5	+1.195	0.2323	+1.544	0.0612
GARCHND log_artg	Parkinson	22	–2.097	0.0360	–1.790	0.9633
GARCHND log_artg	Rogers-Satchell	1	+4.191	0.0000	+4.695	0.0000
GARCHND log_artg	Rogers-Satchell	5	+1.310	0.1903	+1.641	0.0504
GARCHND log_artg	Rogers-Satchell	22	–2.065	0.0389	–1.768	0.9615

Verdict: the rolling-window evidence is markedly more favourable to the GARCHND models than the static-split forecast we ran before. At $h = 1$, both candidates beat the GARCH(1,1) baseline by large, highly-significant margins under both RV proxies (GARCHND tone: DM ≈ +5.4, CW ≈ +5.8, $p < 0.001$; GARCHND log_artg: DM ≈ +4.0, CW ≈ +4.5, $p < 0.001$). Clark–West — the appropriate test for the nested $\gamma = 0$ null — confirms that the news/regime term carries genuine predictive content at the one-day horizon. At $h = 5$ the picture splits: GARCHND tone retains a significant edge (DM p ≈ 0.034–0.038, CW p ≈ 0.01), whereas GARCHND log_artg is no longer significant under DM but is borderline under CW (p ≈ 0.05–0.06). At $h = 22$ the news effect decays: both candidates underperform the baseline (DM < 0; significantly so for log_artg, with one-sided Clark–West $p > 0.95$ rejecting any predictive gain). This is the signature of a short-lived information shock that is correctly priced into next-day variance but adds noise once the iterated forecast horizon exceeds the news half-life. The conclusion overturns the previous "no OOS value" reading: the in-sample γ carries real, horizon-bounded forecasting power, especially when the news regressor is the GDELT tone.

Check 2 — Placebo / regime-only test

Two competing specifications are estimated for each GARCHND κ=50% model: (a) the regime-only model $\gamma \cdot d_3$ (no x^2 term), and (b) five placebos replacing the real x_{t-1} with independent random permutations of it. If the original model does not beat these, the γ estimate reflects regime mean-reversion, not news.

x = tone

Specification	LL	AIC	BIC	γ	SE(γ)	t(γ)
original	-5951.97	11913.94	11943.57	-0.8262	0.0602	-13.72
regime-only ($\gamma \cdot$d3)	-5950.18	11910.37	11939.99	-1.1218	0.0172	-65.10
placebo seed=2026	-5955.76	11921.52	11951.15	-0.1501	0.0283	-5.31
placebo seed=2027	-5949.09	11908.18	11937.80	-0.7548	0.0233	-32.41
placebo seed=2028	-5952.23	11914.46	11944.08	-0.3636	0.0213	-17.03
placebo seed=2029	-5950.81	11911.63	11941.25	-0.8354	0.0175	-47.80
placebo seed=2030	-5952.38	11914.75	11944.38	-0.3935	0.0610	-6.45

Verdict: original does NOT beat the regime-only / placebo baselines (ΔLL vs regime-only = –1.79; placebo LL max = –5949.09 vs original –5951.97). News effect is likely a regime artefact.

x = log_art_growth

Specification	LL	AIC	BIC	γ	SE(γ)	t(γ)
original	-5951.25	11912.49	11942.12	-1.4116	0.2184	-6.46
regime-only ($\gamma \cdot$d3)	-5950.18	11910.37	11939.99	-1.1218	0.0172	-65.10
placebo seed=2026	-5955.15	11920.29	11949.92	-0.8036	0.1258	-6.39
placebo seed=2027	-5955.34	11920.67	11950.30	-0.2475	0.1213	-2.04
placebo seed=2028	-5955.10	11920.19	11949.82	-0.3646	0.1288	-2.83
placebo seed=2029	-5949.72	11909.44	11939.06	-1.0843	0.0814	-13.32
placebo seed=2030	-5954.52	11919.04	11948.66	-0.4900	0.0936	-5.24

Verdict: original does NOT beat the regime-only / placebo baselines (ΔLL vs regime-only = –1.06; placebo LL max = –5949.72 vs original –5951.25). News effect is likely a regime artefact.

Check 3 — SE stability across sigmoid sharpness K

Both GARCHND κ=50% models are refitted at $K \in \{5, 10, 20, 50\}$ and with a hard binary indicator ($K \to \infty$). Stability is measured by the coefficient of variation of γ across K values; $CV < 0.2$ is considered stable.

x = tone (CV = 0.412 — sensitive to K)

K	LL	γ	SE(γ)	t(γ)	converged
K=5	-5952.63	-0.3626	0.0737	-4.92	True
K=10	-5952.42	-0.3769	0.0740	-5.09	True
K=20	-5951.97	-0.8262	0.0602	-13.72	True
K=50	-5952.16	-0.3880	0.0008	-468.3	True
hard d3	-5953.24	-0.3191	0.0989	-3.23	True

x = log_art_growth (CV = 0.119 — stable)

K	LL	γ	SE(γ)	t(γ)	converged
K=5	-5951.98	-1.3781	0.2739	-5.03	True
K=10	-5952.01	-1.0979	0.2493	-4.40	True
K=20	-5951.25	-1.4116	0.2184	-6.46	True
K=50	-5951.42	-1.1276	0.0405	-27.81	True
hard d3	-5951.52	-1.0767	0.0029	-370.8	True

Verdict: the tone specification is sensitive to the sigmoid shape; log_art_growth is stable in sign and magnitude but not in t-ratio (t ranges from −4.4 to −370.8).

Check 4 — High-volatility episode clusters (κ=50%)

Days where $\sigma^2_{t-1} \geq \kappa_{high}$ (annualised 50% vol) are grouped into episodes separated by gaps of at most 5 days. The longest episode determines the recommended block-bootstrap length and HAC bandwidth.

x = tone: 110 triggered days (3.98%), 15 distinct episodes; recommended block length ≥ 36 days

#	Start	End	Days
1	2021-02-10	2021-04-01	36
2	2020-03-11	2020-04-14	24
3	2025-10-15	2025-11-12	21
4	2026-03-05	2026-03-30	18
5	2026-02-03	2026-02-19	12
6	2025-04-11	2025-04-23	8
7	2018-12-24	2019-01-02	6
8	2022-11-08	2022-11-15	6
9	2024-09-30	2024-10-01	2
10	2021-01-11	2021-01-11	1

x = log_art_growth: 129 triggered days (4.67%), 22 distinct episodes; recommended block length ≥ 32 days

#	Start	End	Days
1	2020-03-04	2020-04-17	32
2	2021-01-07	2021-02-18	29
3	2021-03-01	2021-04-01	24
4	2025-10-15	2025-10-29	11
5	2024-09-30	2024-10-11	10
6	2026-02-03	2026-02-17	10
7	2022-05-11	2022-05-23	9
8	2018-12-24	2019-01-03	7
9	2025-04-11	2025-04-22	7
10	2026-03-24	2026-03-31	6

Check 5 — HAC and Moving-Block-Bootstrap SEs on γ (Notebook 07)

Sandwich (HAC) and moving-block-bootstrap (MBB, B=2000) standard errors on γ for both GARCHND κ=50% specifications, at block lengths L ∊ {25, 50, 75, 100} days (informed by the episode clusters in Check 4).

x = tone

SE method	γ	SE(γ)	t(γ)
iid sandwich	-0.8262	0.0602	-13.72
HAC L=25	-0.8262	0.0586	-14.09
HAC L=50	-0.8262	0.0548	-15.07
HAC L=75	-0.8262	0.0565	-14.62
HAC L=100	-0.8262	0.0578	-14.30
MBB L=25	-0.8262	0.0583	-14.18
MBB L=50	-0.8262	0.0563	-14.68
MBB L=75	-0.8262	0.0575	-14.36
MBB L=100	-0.8262	0.0596	-13.87

x = log_art_growth

SE method	γ	SE(γ)	t(γ)
iid sandwich	-1.4116	0.2184	-6.46
HAC L=25	-1.4116	0.2321	-6.08
HAC L=50	-1.4116	0.2424	-5.82
HAC L=75	-1.4116	0.2469	-5.72
HAC L=100	-1.4116	0.2473	-5.71
MBB L=25	-1.4116	0.2360	-5.98
MBB L=50	-1.4116	0.2455	-5.75
MBB L=75	-1.4116	0.2471	-5.71
MBB L=100	-1.4116	0.2503	-5.64

Verdict: HAC and MBB corrections do NOT inflate the standard errors materially. The t-ratios remain strongly significant (tone: around –14; log_artg: around –5.7). This is explained by the iid nature of x^2_{t-1} within the regime, which absorbs serial correlation in the score. The conclusion from Checks 1–2 (regime artefact, no OOS gain) stands unaffected.

Check 6 — GDELT coverage of REMX / rare earths (Notebook 07)

The GDELT article cache (314,048 articles, 2015-04-01 to 2026-04-01) was built with a permissive 19-concept filter (any single match passes). This check quantifies how many articles are genuinely related to the REMX economy across four progressively tighter filter tiers.

Articles per concept (full cache)

Concept	N articles	Share
has_cerium	186,346	59.34%
has_rare_earth	92,383	29.42%
has_lynas	26,066	8.30%
has_samarium	5,005	1.59%
has_neodymium	2,960	0.94%
has_mp_materials	2,891	0.92%
has_praseodymium	2,124	0.68%
has_arafura	1,730	0.55%
has_europium	1,153	0.37%
has_terbium	1,143	0.36%
has_gadolinium	1,051	0.33%
has_yttrium	1,030	0.33%
has_lanthanum	877	0.28%
has_china_northern	702	0.22%
has_dysprosium	576	0.18%
has_ndfeb	505	0.16%
has_ndpr	454	0.14%
has_remx	150	0.05%
has_jl_mag	28	0.01%

Filter tiers

Tier	N articles	Art/day	% zero-days	Mean tone	Std tone
Broad (≥1 concept, used in model)	314,048	78.5	0.5%	–0.280	3.299
Medium (metal OR company)	231,708	57.9	0.5%	–0.453	3.585
Tight (rare_earth AND metal/co.)	10,158	2.5	38.5%	–0.130	2.133
Ultra-tight (metal AND company)	559	0.14	91.4%	+0.638	1.850